



KARNATAKA SSLC PATTERN - PRACTICE QUESTION PAPER

S.S.L.C. / CLASS X - MATHEMATICS

English Medium | Subject Code: 81-E

Time: 3 Hours 15 Minutes	Max. Marks: 80
Portion: Chapters 1-5 only	Total Questions: 38

Portion: 1. Real Numbers 2. Polynomials 3. Pair of Linear Equations in Two Variables 4. Quadratic Equations 5. Arithmetic Progressions

General Instructions to the Candidate:

1. This question paper consists of 38 questions.
2. Follow the instructions given against each section and question.
3. Figures in the right-hand margin indicate the maximum marks for the questions.
4. The maximum time includes 15 minutes for reading the question paper.
5. All questions are based only on the first five chapters listed above.

I. Four alternatives are given for each of the following questions / incomplete statements. Choose the correct alternative and write the complete answer along with its letter of alphabet. **8 × 1 = 8**

1. If $HCF(84, 126) = 42$, then $LCM(84, 126)$ is
(A) 126 (B) 168 (C) 252 (D) 336
2. The degree of the polynomial $p(x) = 5x^3 - 2x + 7$ is
(A) 1 (B) 2 (C) 3 (D) 5
3. The pair of equations $2x + 3y = 5$ and $4x + 6y = 10$ has
(A) no solution (B) one solution (C) infinitely many solutions (D) exactly two solutions
4. The roots of $x^2 - 5x + 6 = 0$ are
(A) 1, 6 (B) 2, 3 (C) -2, -3 (D) 3, 6
5. The common difference of the AP 7, 11, 15, 19, ... is
(A) 2 (B) 3 (C) 4 (D) 5
6. If $p(x) = x^2 - 9$, then the sum of its zeroes is
(A) -9 (B) 0 (C) 9 (D) 1
7. For the pair $ax + by + c = 0$ and $a_2x + b_2y + c_2 = 0$, the condition for a unique solution is
(A) $a/a_2 = b/b_2 = c/c_2$ (B) $a/a_2 = b/b_2 \neq c/c_2$ (C) $a/a_2 \neq b/b_2$ (D) $c = 0$
8. The n th term of an AP with first term a and common difference d is
(A) $a + nd$ (B) $a + (n - 1)d$ (C) $n(a + d)$ (D) $a(n - d)$

II. Answer the following questions. **8 × 1 = 8**

9. Use Euclid's division algorithm to find the HCF of 135 and 225.
10. Find the zeroes of the polynomial $x^2 - 7x + 12$.

11. Determine whether the pair $3x + 2y = 5$ and $6x + 4y = 12$ is consistent or inconsistent.
12. Find the discriminant of $2x^2 - 7x + 3 = 0$.
13. Find the 10th term of the AP 4, 9, 14, 19, ...
14. If the zeroes of $x^2 + kx + 12$ are 3 and 4, find k.
15. Solve by substitution: $x + y = 9$ and $x - y = 3$.
16. Find the sum of the first 12 terms of the AP 5, 8, 11, 14, ...

III. Answer the following questions.

$8 \times 2 = 16$

17. Prove that $\sqrt{5}$ is an irrational number.
18. Find the zeroes of $2x^2 - 5x - 3$ and verify the relationship between the zeroes and the coefficients.
19. Solve the pair of linear equations by elimination method: $3x + 2y = 12$ and $5x - 2y = 4$.
20. Find the value of k for which the pair $2x + 3y = 7$ and $4x + ky = 14$ has infinitely many solutions.
21. Solve the quadratic equation $x^2 - 7x + 10 = 0$ by factorisation method.
22. Solve $3x^2 - 8x + 4 = 0$ using the quadratic formula.
23. Which term of the AP 21, 18, 15, ... is -81 ?
24. The 7th term of an AP is 22 and the 13th term is 40. Find its first term and common difference.

IV. Answer the following questions.

$9 \times 3 = 27$

25. Using Euclid's division algorithm, find the HCF of 867 and 255. Hence, express the HCF as a linear combination of 867 and 255.
26. If α and β are the zeroes of $3x^2 - 7x + 2$, form a quadratic polynomial whose zeroes are 2α and 2β .
27. Solve by cross-multiplication method: $2x + 3y = 13$ and $4x - y = 5$.
28. Solve graphically the pair of equations $x + y = 6$ and $x - y = 2$. Write the coordinates of the point of intersection.
29. The product of two consecutive positive integers is 306. Form the corresponding quadratic equation and find the integers.
30. Find the nature of the roots of $5x^2 - 6x + 2 = 0$ using the discriminant. State whether the roots are real and distinct, real and equal, or not real.
31. Find the sum of all multiples of 7 between 50 and 200 using the formula for the sum of an AP.
32. The 5th term of an AP is 18 and the 11th term is 42. Find the first term, common difference and 20th term.
33. A student saves ₹200 in the first month and increases the saving by ₹50 every month. Find the total amount saved in 12 months and the amount saved in the 12th month.

V. Answer the following questions.

$4 \times 4 = 16$

34. Prove that the square of any odd positive integer is of the form $8m + 1$, where m is an integer.
35. A pair of linear equations is given by $3x + 4y = 10$ and $6x + 8y = 20$. Represent the equations graphically and state the nature and number of solutions. OR Solve the same pair using any algebraic method and state the nature of the solution.
36. The sum of the first 10 terms of an AP is 155 and its 10th term is 29. Find the first term and common difference.

37. The area of a rectangular garden is 300 m^2 . Its length is 5 m more than its breadth. Form a quadratic equation and find the dimensions of the garden.

VI. Answer the following question.

$1 \times 5 = 5$

38. A theatre has 20 seats in the first row, 24 seats in the second row, 28 seats in the third row and so on. There are 25 rows. Find: (a) the number of seats in the last row, and (b) the total number of seats in the theatre. If each row must have 4 more seats than the previous row, explain why the sequence forms an AP.